Returns to Schooling in Brazil, 2012–2025: Design-Based Mincer Estimates from the PNAD Contínua

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Abstract

Brazil added almost two years of schooling to its workforce between 2012 and 2025. This paper asks what a year of schooling is worth in the labour market over that period, and for whom. Using the first-visit microdata of the PNAD Contínua for 2012–2019 and 2022–2025, 1.77 million workers aged 25–64, I estimate Mincer earnings equations with standard errors that respect the survey's strata, clusters and weights, after first reproducing IBGE's own published aggregates from the microdata. The return to a year of schooling is 9.8 percent and varies little across years, between 9.3 and 10.2 percent. That stable average hides a convex profile and a steady compression of the premia attached to each completed level: the premium of a university degree over no schooling fell from 1.62 to 1.29 log points. Returns are higher for women, lower for Black and Brown workers and highest in the Nordeste. Unweighted OLS with robust standard errors understates uncertainty threefold. The estimates are the descriptive benchmark that Duflo (2001) compares with her instrumental-variables results; they are not causal effects.

Resumo. O Brasil acrescentou quase dois anos de estudo à sua força de trabalho entre 2012 e 2025. Este artigo pergunta quanto vale um ano de estudo no mercado de trabalho nesse período, e para quem. Com os microdados da primeira visita da PNAD Contínua de 2012–2019 e 2022–2025, 1,77 milhão de trabalhadores de 25 a 64 anos, estimo equações de Mincer com erros-padrão que respeitam os estratos, os conglomerados e os pesos da pesquisa, depois de reproduzir os agregados oficiais do IBGE a partir dos microdados. O retorno a um ano de estudo é de 9,8% e varia pouco entre os anos, de 9,3% a 10,2%. Essa média estável esconde um perfil convexo e uma compressão contínua dos prêmios de cada nível concluído: o prêmio do ensino superior sobre a ausência de instrução caiu de 1,62 para 1,29 pontos de log. Os retornos são maiores para mulheres, menores para pretos e pardos e mais altos no Nordeste. MQO sem pesos e com erros-padrão robustos subestima a incerteza em três vezes. As estimativas são o padrão descritivo com que Duflo (2001) compara seus resultados por variáveis instrumentais; não são efeitos causais.

JEL: I26, J31, O15, O54. **Keywords:** returns to schooling; Mincer equation; Brazil; PNAD Contínua; complex survey design; replication.

1 Introduction

Few numbers in economics are asked for as often as the return to schooling: the extra earnings that one more year in school buys. Mincer (1974) gave the question its standard form, a regression of log earnings on years of schooling and experience, and the coefficient on schooling has since been estimated for most countries in the world. Returns are typically higher in poorer countries and in Latin America than in rich ones, and they tend to fall as schooling spreads (Psacharopoulos 1994; Psacharopoulos and Patrinos 2018). Brazil is a case in point. In the 1990s its return was among the highest measured anywhere, and the Brazilian literature has documented both its decline and its unusual shape, with much of the pay-off concentrated at the top of the schooling distribution (Lam and Schoeni 1993; Menezes-Filho 2001; Barbosa Filho and Pessôa 2008; Menezes-Filho et al. 2006; Ferreira et al. 2022). Since then the country has kept adding schooling at a fast pace. Among occupied workers aged 25–64, mean schooling rose from 9.8 years in 2012 to 11.6 years in 2025. What happened to the return while the supply of educated workers grew that quickly is the question of this paper.

The method is deliberately plain. I estimate the Mincer equation on the annual first-visit microdata of the PNAD Contínua, the continuous household survey of the Brazilian statistical office (IBGE), for every year released between 2012 and 2025, which excludes 2020 and 2021. Two features distinguish the exercise from a textbook regression. First, the standard errors respect how the survey was drawn. The PNAD Contínua does not sample people at random across the country: it first selects census tracts, then households inside them, and it weights the result to match the population. People in the same tract are more alike than people chosen at random, so a sample of 150,000 workers carries less information than its size suggests, and standard errors that ignore this are too small (Lumley 2004). Second, before any regression is run, the code reproduces IBGE's own published figures for mean labour income and mean schooling from the same microdata. If the weights, the deflators or the sample definitions were wrong, that check would fail. It passes to within 0.01 percent of the published labour-income series in eleven of twelve years.

The exercise belongs to a series that re-runs the empirical methods of the MIT MicroMasters course *Foundations of Development Policy* on Brazilian data. The Week 3 paper of that course is Duflo (2001), which uses a school-construction program in Indonesia to estimate a causal return to schooling and compares it with the ordinary least-squares (OLS) return of 7.8 percent in the same data. The comparison is the point: OLS measures how much more educated people earn, not how much more a given person would earn with another year of school, because the two differ in ability, family background and the schools they attended (Griliches 1977; Card 1999). This paper does the OLS half of that comparison for Brazil, with the care that the survey deserves, and leaves the causal half, which needs a natural experiment, to a later paper in the series.

The results are easy to summarise. Pooled over twelve survey years, a year of schooling is associated with 9.8% higher hourly earnings, and the year-by-year estimates move only between 9.3% and 10.2%, with the lowest values after the pandemic. That stability is misleading on its own. The relation between schooling and earnings is strongly convex, with an implied 20% per year of university against 5% per year of primary school, and every level premium fell over the

period: relative to workers with no schooling, a completed degree was worth 1.62 log points in 2012 and 1.29 in 2025. A single Mincer coefficient can stay flat while all premia fall if the workforce moves up a convex profile, which is what happened. The gradient is steeper for women than for men, flatter for Black and Brown workers than for White and Asian workers, so that the gender gap narrows with schooling while the racial gap widens, and it is steepest in the Nordeste. Finally, treating the survey as a random sample and using robust standard errors makes the estimates look 3.4 times more precise than they are.

Section 2 explains what Duflo's paper and its recent reanalysis teach about doing this well. Sections 3 and 4 describe the data and the method. Sections 5 and 6 present the results and their robustness, Section 7 what the numbers cannot tell us, and Section 8 concludes.

2 What Duflo (2001) and its reanalysis teach

Between 1973 and 1978 Indonesia built more than 61,000 primary schools, about two for every thousand children, and the number of schools each district received was set by a rule: proportional to the number of children not in school in 1972. Duflo (2001) noticed that this created two sources of variation that an adult cannot manipulate. Children who were twelve or older in 1974 had already finished primary school and could not benefit; younger children could, and more so where more schools were built. Comparing the schooling and the 1995 wages of young and old cohorts across districts with more and fewer schools, she finds that each school per thousand children raised schooling by 0.12–0.19 years and wages by 1.5–2.7 percent. Dividing the wage effect by the schooling effect gives a causal return to a year of schooling between 6.8 and 10.6 percent, close to the 7.8 percent that a plain Mincer regression gives in the same sample. Her reading was that in Indonesia the OLS return was not inflated by family background, and that it was lower than in earlier decades because schooling had become much more common, a supply effect of the kind described by Katz and Murphy (1992).

The paper has been revisited. Roodman (2022) rebuilt the data from primary sources, corrected entry errors in about a tenth of the observations, clustered the standard errors by district, the unit at which schools were assigned (Bertrand et al. 2004), followed the cohorts into later surveys, and applied modern tests for weak instruments. The headline estimates do not survive all of these changes, and the causal return turns out to be too imprecise to pin down. The patterns that made the original argument, effects appearing only in the exposed cohorts and at the grade levels where schools were built, partly persist. He also found that the original code weighted some estimates and not others, a choice the paper never discussed.

Three habits follow, and this paper adopts them. State every inference choice in the text rather than leaving it to the code; keep the same controls in every table; and report the shape of the relationship, not only one coefficient, because patterns are what survive scrutiny. The convexity of the Brazilian profile and the jumps at completed levels, which Hungerford and Solon (1987) called sheepskin effects, are the Brazilian counterpart of Duflo's cohort fingerprints.

3 Data

3.1 The PNAD Contínua

The PNAD Contínua has been in the field since 2012 and interviews about 211,000 households every quarter (IBGE 2024). Its sample is drawn in two stages: census tracts are selected within strata that combine state, metropolitan status and other criteria, and households are then selected within each tract. Every household is interviewed five times over five consecutive quarters. IBGE publishes the first of these interviews as an annual file, which contains the full questionnaire on schooling and work, and it is this first-visit file that IBGE uses for its own annual labour-income releases. I use every first-visit file released: 2012 to 2019 and 2022 to 2025. During the pandemic fieldwork moved to the telephone and IBGE did not publish first-visit files for 2020 and 2021, so those two years are absent from every figure and table. The files were downloaded on 11 October 2026 with the PNADcIBGE package (Braga and Assunção 2024), which also attaches the deflators IBGE publishes for them; earnings are expressed at 2025 average prices throughout.

Three variables describe the design: the stratum, the census tract, which is the primary sampling unit, and the first-visit weight, which IBGE calibrates so that the weighted sample matches population projections by age, sex and area. All three enter the analysis.

3.2 Sample and variables

The dependent variable is the log of real hourly earnings in the main job, computed as habitual monthly earnings divided by habitual weekly hours times 52/12. Schooling is IBGE's standardised count of completed years, from 0 to 16, in which 9 years means a completed *ensino fundamental*, 12 a completed *ensino médio* and 16 a completed university degree; the companion variable records the highest level completed. The controls are age and its square, sex, race in IBGE's categories grouped as White or Asian, Black or Brown, and Indigenous, an urban indicator and state fixed effects; the pooled specifications add year fixed effects.

The analysis sample is every occupied person aged 25 to 64 with positive habitual earnings and between 1 and 120 weekly hours in the main job, with schooling and race reported. Unlike Duflo's Indonesian survey, which recorded wages for employees only, the PNAD Contínua records earnings for the self-employed and for employers, so they are included; a sample of wage employees only is reported as a robustness check. Table 7 in the appendix gives the count at each filter step and year. Between 95.8% and 98.4% of occupied people in the age range make it into the sample; the main loss is unpaid family workers, who have no earnings. The final sample has 1,773,567 person-years.

Table 1 shows what the sample looks like year by year. Mean schooling rose by almost two years; real hourly earnings fell during the 2015–2016 recession and recovered after 2022, ending at R$ 22.29 an hour at 2025 prices; the share of women rose from 41.4% to 43.7% and the share of Black and Brown workers from 50.1% to 54.7%; the share in formal work stayed near 64%.

Table 1: Analysis sample by year: design-based means

Year	Observations	Schooling	Hourly earnings	Age	Women	Black or Brown	Formal
2012	146,662	9.84	19.62	40.1	0.414	0.501	0.632
2013	152,993	9.96	20.28	40.2	0.417	0.506	0.642
2014	156,830	10.12	20.72	40.4	0.419	0.508	0.654
2015	155,348	10.27	20.13	40.6	0.421	0.516	0.656
2016	154,607	10.42	19.92	40.7	0.424	0.522	0.649
2017	152,398	10.56	19.91	40.8	0.426	0.528	0.634
2018	151,572	10.78	20.09	41.0	0.430	0.535	0.626
2019	150,773	10.88	20.04	41.1	0.431	0.540	0.625
2022	130,659	11.19	19.60	41.3	0.431	0.539	0.623
2023	133,978	11.30	20.41	41.5	0.434	0.544	0.625
2024	137,271	11.40	21.06	41.6	0.436	0.553	0.625
2025	150,476	11.59	22.29	41.7	0.437	0.547	0.633

Occupied persons aged 25-64 with positive habitual earnings and 1-120 weekly hours in the main job. Means use the first-visit weights. Schooling in years; hourly earnings in the main job at 2025 prices (R$); women, Black or Brown and formal are shares. Formal: signed-card or statutory employees, or self-employed and employers contributing to social security. Weighted sample sizes range from 68.5 million (2012) to 83.7 million (2025) person-years.

3.3 Checking the microdata against IBGE's published figures

Weights, deflators and the exact definition of who counts as "occupied with earnings" are where survey analyses most often go wrong, and the errors are invisible in a regression table. So before any model is estimated, the code recomputes two series that IBGE publishes from these same files and compares them with the published values (Table 8 in the appendix). Mean real monthly labour income of occupied persons aged 14 or more, habitually and effectively received, is reproduced from SIDRA table 7445 to within 0.01 percent in every year except 2022, where the current microdata file gives values 3.3 percent above the published ones; the published 2022 figures appear to predate the 2025 revision of the microdata, and the gap is recorded rather than hidden. Mean years of schooling of persons aged 25 or more, in total and by sex, is reproduced from SIDRA table 7126 to within 1.0%; that series comes from a different quarter's supplement and is published rounded to one decimal, so a closer match is not expected. The exercise also pinned down one definitional detail: IBGE's published mean of effective earnings averages over everyone with positive habitual earnings, including months with zero effective earnings.

4 Method

4.1 The earnings equation

The baseline is Mincer's equation,

$$\ln w_i = \beta S_i + \gamma_1 \text{age}_i + \gamma_2 \text{age}_i^2 + \mathbf{x}_i'\delta + \alpha_{s(i)} + \varepsilon_i, \tag{1}$$

where w_i is the hourly earnings of worker i, S_i her years of schooling, $\mathbf{x}_i$ the sex, race and urban indicators and $\alpha_{s(i)}$ a fixed effect for her state. The coefficient β is the proportional difference in hourly earnings between two workers who differ by one year of schooling and share the same age, sex, race, area and state. Mincer (1974) derived the equation from a model in which schooling is an investment that pays a constant rate of return, which is why the dependent variable is in logs and why β is read as a rate; Heckman et al. (2006) review what the derivation assumes and how the equation has held up. Mincer used potential experience, age minus schooling minus six, rather than age; I follow Lemieux (2006) in preferring an age polynomial, because potential experience is itself a function of schooling and the estimate of β then depends on how that function is specified.

The equation is estimated separately for each survey year, which traces the trend in β, and pooled over all years with year fixed effects. Two variants loosen the assumption of a constant rate. The first replaces years of schooling by indicators for the highest level completed, so that the return can differ along the schooling distribution and can jump at the completion of a level. Hungerford and Solon (1987) and Jaeger and Page (1996) showed for the United States that such jumps exist, which they attributed to the diploma signalling something beyond the years it took; the Brazilian literature finds them large (Barbosa Filho and Pessôa 2008). The second interacts years of schooling with sex and with race, so that each sex-by-race group has its own slope; the slope for a group is a sum of coefficients and its standard error comes from the full covariance matrix. Regional estimates re-run Equation 1 within each of the five macro-regions.

4.2 Inference

Standard errors come from the design of the survey, in the sense of Lumley (2004) and Lumley (2010). For each year the design object is built from the full first-visit file, with the census tract as the sampling unit, the stratum as the stratum and the first-visit weight as the weight, and the analysis sample is then taken as a subset of that design, so that the variance formula keeps the right count of sampling units per stratum. The coefficients are weighted least squares; their variance is estimated by linearisation, which asks how much the estimate would change if each census tract were dropped, and the reference distribution is a t with degrees of freedom equal to the number of tracts minus the number of strata, about 44,580 in the pooled design. The pooled design stacks all years and treats the same tract in different years as one cluster, which is the conservative choice for a survey that reuses its master sample.

Two shortcuts that practitioners often take are reported for comparison: unweighted OLS with heteroskedasticity-robust (HC1) standard errors, which treats every worker as an independent

draw, and weighted OLS with standard errors clustered by tract, which keeps the clustering but ignores the stratification. Bertrand et al. (2004) showed how badly inference goes when clustering is ignored in panel data; Solon et al. (2015) discuss when weights change the estimate and when they only change its precision; Abadie et al. (2023) explain which level to cluster at. The comparison in Section 6 shows what each layer of the design contributes here.

5 Results

5.1 The return to a year of schooling, 2012–2025

Table 3 reports the pooled specifications and Table 2 and Figure 1 the year-by-year estimates of Equation 1. Pooled over the twelve survey years, a year of schooling is associated with 9.8% higher hourly earnings (standard error 0.05%, 1,773,567 observations). Year by year, the coefficient starts at 9.9% in 2012, drifts down to 9.7% in 2016, climbs to 10.2% in 2019, and is lowest in the three post-pandemic years, at 9.4%, 9.3% and 9.3%, before returning to 9.8% in 2025. With standard errors of about 0.10% points, most of the year-to-year movement is within sampling error; the drop from 2018–2019 to 2022–2024 is not, since the confidence intervals [10.0%, 10.4%] and [9.1%, 9.5%] do not overlap. The Brazilian number sits above the 7.8 percent that Duflo (2001) finds for Indonesian wage earners in 1995 and in the upper part of the Latin American range in Psacharopoulos and Patrinos (2018), but well below the 15 percent or more that Lam and Schoeni (1993) and Menezes-Filho (2001) report for Brazil in the 1980s and 1990s, when secondary and university graduates were far scarcer.

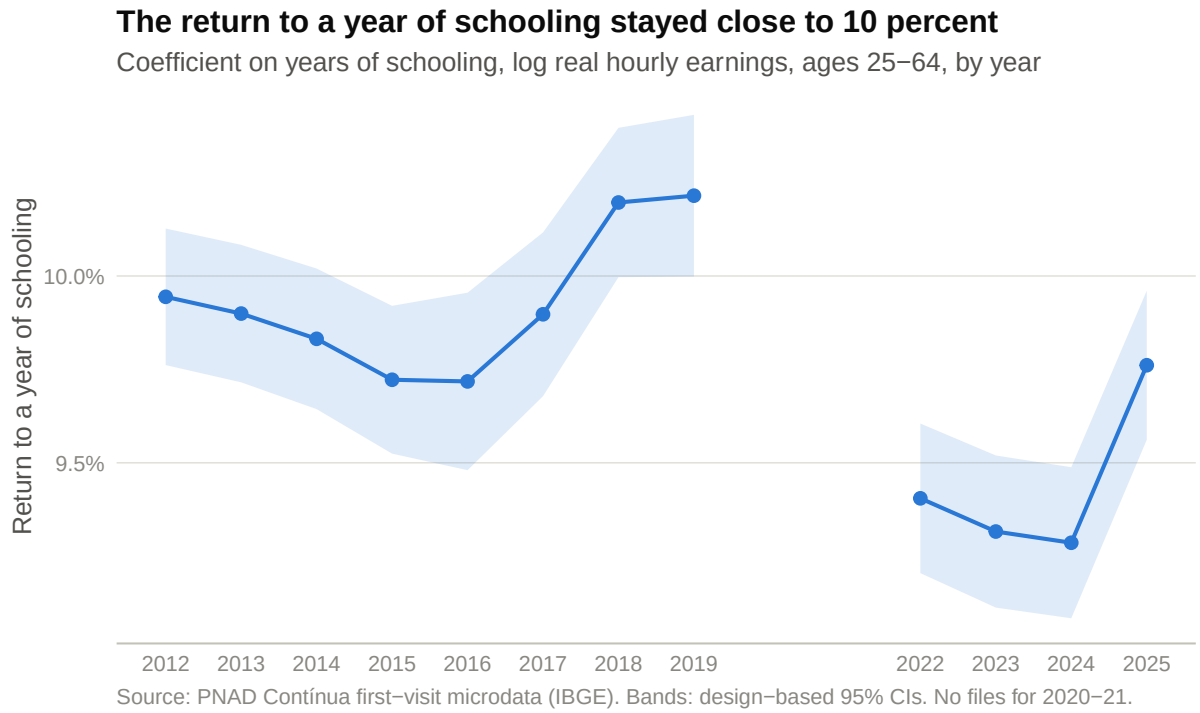


Figure 1: The return to a year of schooling by survey year. Coefficient on years of schooling in the linear specification with state fixed effects, one regression per year, workers aged 25–64. Shaded bands are design-based 95 percent confidence intervals. No first-visit microdata were released for 2020 and 2021.

Table 2: Return to a year of schooling by survey year

Year	Estimate	SE	95% CI	Observations
2012	0.0994	(0.0009)	[0.098, 0.101]	146,662
2013	0.0990	(0.0009)	[0.097, 0.101]	152,993
2014	0.0983	(0.0010)	[0.096, 0.100]	156,830
2015	0.0972	(0.0010)	[0.095, 0.099]	155,348
2016	0.0972	(0.0012)	[0.095, 0.100]	154,607
2017	0.0990	(0.0011)	[0.097, 0.101]	152,398
2018	0.1020	(0.0010)	[0.100, 0.104]	151,572
2019	0.1021	(0.0011)	[0.100, 0.104]	150,773
2022	0.0940	(0.0010)	[0.092, 0.096]	130,659
2023	0.0932	(0.0010)	[0.091, 0.095]	133,978
2024	0.0929	(0.0010)	[0.091, 0.095]	137,271
2025	0.0976	(0.0010)	[0.096, 0.100]	150,476

Design-based standard errors in parentheses, computed from the PNAD Contínua strata, primary sampling units and first-visit weights (V1032). Dependent variable: log real hourly earnings in the main job (habitual monthly earnings at 2025 prices divided by 52/12 times weekly hours). Sample: occupied persons aged 25-64 with positive earnings and 1-120 weekly hours. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Coefficient on years of schooling in the linear specification with state fixed effects, one regression per year. No first-visit microdata were released for 2020 and 2021.

Table 3: Mincer regressions, pooled 2012-2025

	(1) Linear	(2) Levels	(3) Interactions
Years of schooling	0.0976***		0.1080***
	(0.0005)		(0.0008)
Fundamental incomplete		0.2455***	
		(0.0050)	
Fundamental complete		0.4248***	
		(0.0055)	
Médio incomplete		0.4794***	
		(0.0058)	
Médio complete		0.6340***	
		(0.0054)	
Superior incomplete		0.9275***	
		(0.0068)	
Superior complete		1.4470***	
		(0.0072)	

	(1) Linear	(2) Levels	(3) Interactions
Schooling × Women			0.0092***
			(0.0004)
Schooling × Black/Brown			-0.0251***
			(0.0007)
Schooling × Indigenous			-0.0292***
			(0.0039)
Women	-0.2510***	-0.2639***	-0.3518***
	(0.0016)	(0.0015)	(0.0042)
Black/Brown	-0.1532***	-0.1177***	0.1208***
	(0.0025)	(0.0021)	(0.0062)
Indigenous	-0.1643***	-0.1384***	0.1511***
	(0.0151)	(0.0150)	(0.0386)
Age	0.0463***	0.0414***	0.0461***
	(0.0007)	(0.0007)	(0.0007)
Age squared	-0.0004***	-0.0004***	-0.0004***
	(0.0000)	(0.0000)	(0.0000)
Urban	0.2020***	0.2592***	0.2106***
	(0.0036)	(0.0037)	(0.0036)
Observations	1,773,567	1,773,567	1,773,567
State and year fixed effects	Yes	Yes	Yes

Design-based standard errors in parentheses, computed from the PNAD Contínua strata, primary sampling units and first-visit weights (V1032). Dependent variable: log real hourly earnings in the main job (habitual monthly earnings at 2025 prices divided by 52/12 times weekly hours). Sample: occupied persons aged 25-64 with positive earnings and 1-120 weekly hours. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Education levels are relative to no schooling; interactions are relative to men and to White/Asian workers.

The other coefficients behave as the literature leads one to expect and are stable across specifications. Holding schooling, age, area and state constant, women earn 25% log points less per hour than men and Black and Brown workers 15% log points less than White and Asian workers; urban workers earn 20% log points more than rural ones; earnings peak at 59 years of age.

5.2 Premia by completed level

Column (2) of Table 3 replaces years of schooling by the level completed, and Table 4 and Figure 2 follow the level premia over time. The profile is strongly convex. Relative to workers with no schooling, a completed *ensino fundamental* is worth 0.42 log points, or 4.7% for each of its nine years; a completed *ensino médio* adds 0.21 log points over three more years, 7.0% a year; and a completed degree adds 0.81 log points over four more years, 20.3% a year. Finishing a level is worth more than the years it takes. The step from an incomplete to a complete *fundamental* is 0.18 log points, from an incomplete to a complete *médio* 0.15, and from an incomplete to a

complete degree 0.52, sheepskin effects of the kind documented by Hungerford and Solon (1987) and far larger than those they found.

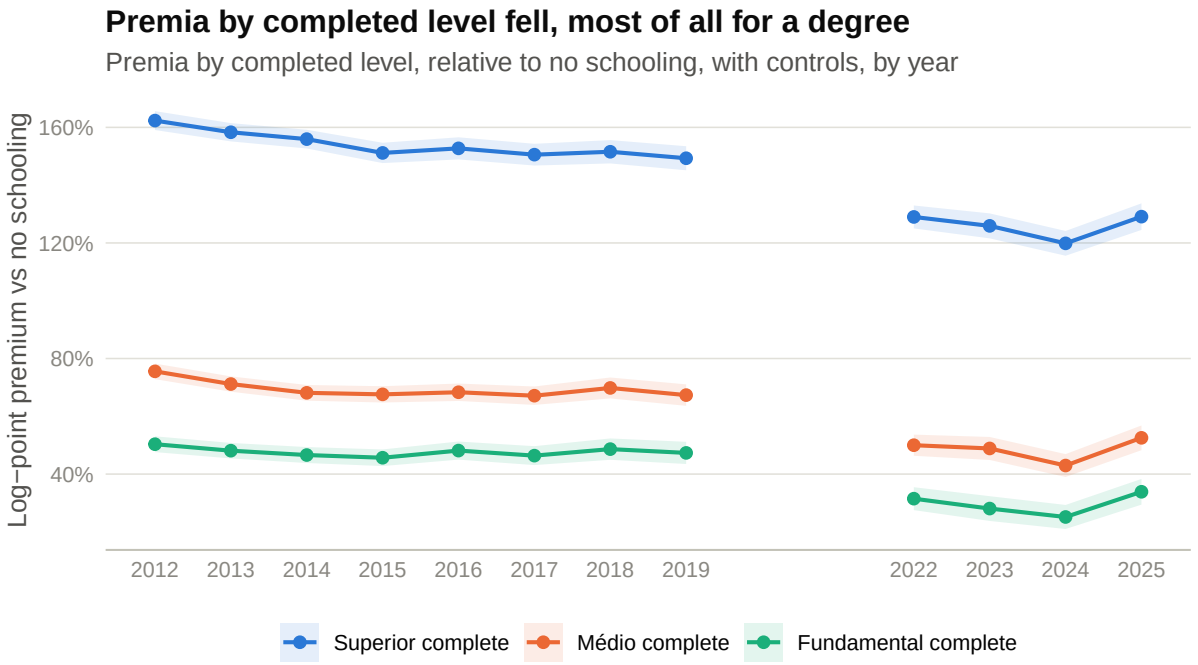


Figure 2: Earnings premia by completed level of education, relative to no schooling, with the same controls as the linear specification; one regression per year. Shaded bands are design-based 95 percent confidence intervals.

Table 4: Earnings premia by completed level of education, relative to no schooling

	2012	2019	2025	Pooled
Fundamental incomplete	0.315***	0.286***	0.187***	0.246***
	(0.013)	(0.018)	(0.022)	(0.005)
Fundamental complete	0.503***	0.473***	0.339***	0.425***
	(0.014)	(0.020)	(0.023)	(0.005)
Médio incomplete	0.579***	0.508***	0.378***	0.479***
	(0.016)	(0.021)	(0.023)	(0.006)
Médio complete	0.756***	0.673***	0.526***	0.634***
	(0.014)	(0.019)	(0.022)	(0.005)
Superior incomplete	1.133***	0.956***	0.792***	0.928***
	(0.018)	(0.023)	(0.025)	(0.007)
Superior complete	1.623***	1.493***	1.291***	1.447***
	(0.017)	(0.021)	(0.023)	(0.007)
Observations	146,662	150,773	150,476	1,773,567

	2012	2019	2025	Pooled

Design-based standard errors in parentheses, computed from the PNAD Contínua strata, primary sampling units and first-visit weights (V1032). Dependent variable: log real hourly earnings in the main job (habitual monthly earnings at 2025 prices divided by 52/12 times weekly hours). Sample: occupied persons aged 25-64 with positive earnings and 1-120 weekly hours. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Controls: age, age squared, sex, race, urban, state fixed effects (and year fixed effects in the pooled column).

Every premium fell. Relative to no schooling, a completed degree was worth 1.62 log points in 2012, 1.49 in 2019 and 1.29 in 2025; a completed *médio* went from 0.76 to 0.53 and a completed *fundamental* from 0.50 to 0.34. Because the decline is proportionally similar at every level, it is unlikely to be a story about university graduates alone. Two things could be going on. The reference group of workers with no schooling shrank and aged while the real minimum wage rose through the 2010s, which lifted the bottom of the earnings distribution; Engbom and Moser (2022) show that the minimum wage explains a large part of the compression of Brazilian earnings in these years. And the supply of workers at every completed level kept expanding, which in the supply-and-demand framework of Katz and Murphy (1992) lowers the price of each level.

How can every level premium fall while the single Mincer coefficient barely moves? The linear coefficient is not a separate fact; it is a weighted average of the slopes between adjacent levels, with weights that depend on where the workers are in the schooling distribution (Yitzhaki 1996; Angrist and Krueger 1999). As mean schooling rose by almost two years, more of the workforce sat on the steep segment between *médio* and university, and that segment received more weight. Menezes-Filho et al. (2006) described the same composition mechanism for Brazilian inequality a decade earlier. The arithmetic fits the data, but it is an interpretation; the paper does not decompose the change formally.

5.3 Sex and race

Column (3) of Table 3 lets the slope differ by sex and race, and Figure 3 traces the four resulting gradients year by year. Pooled over all years, a year of schooling is worth 10.8% for White and Asian men, 11.7% for White and Asian women, 8.3% for Black and Brown men and 9.2% for Black and Brown women, each with a standard error below 0.08% points. Two conditional gaps follow. The gender gap narrows with schooling: at nine years women earn 27% log points less than men with the same characteristics, at sixteen years 20% log points less. The racial gap widens with schooling: it is 10% log points at nine years, 18% at twelve and 28% at sixteen. The ordering of the four gradients is the same in every year, and all four dip after the pandemic.

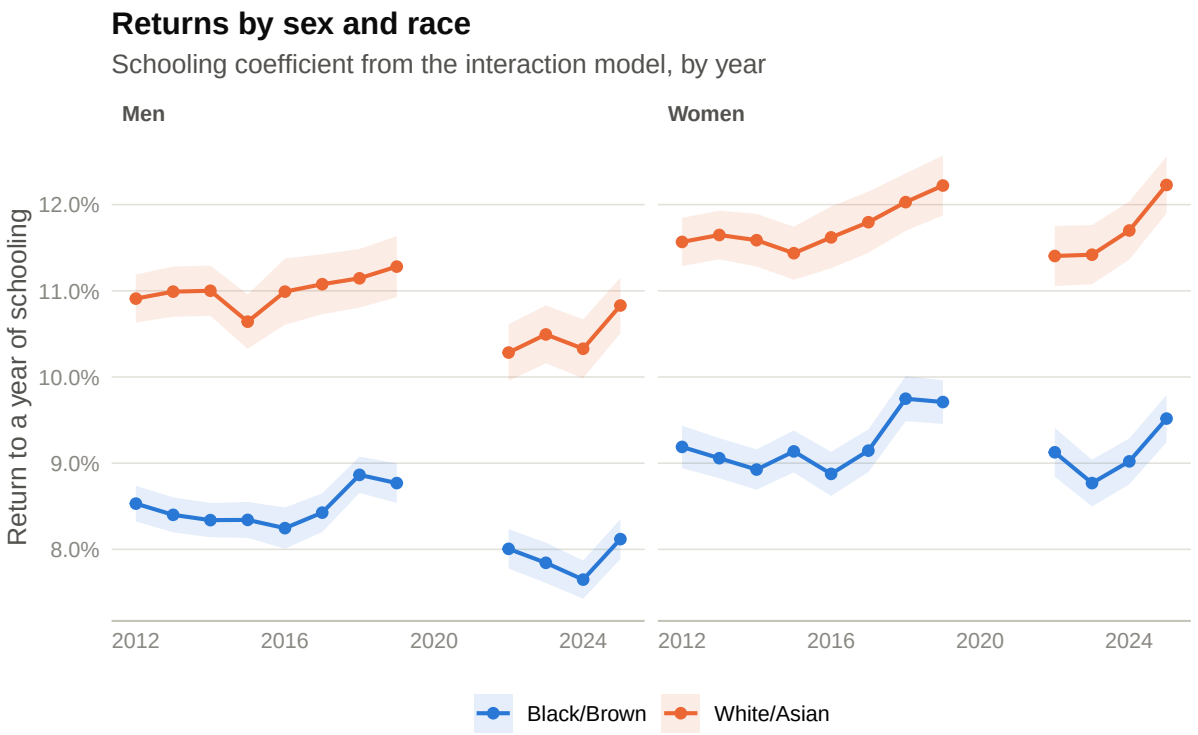


Figure 3: Return to a year of schooling by sex and race. Group-specific schooling coefficients from the interaction specification, one regression per year, with design-based 95 percent confidence intervals.

These are differences among people who work. Who works differs by sex and race, and earnings are observed only for them, so a steeper gradient for women can reflect the stronger selection of less-educated women out of paid work as much as a higher pay-off to schooling, the problem analysed by Heckman (1979). The gradients are reported because they are precisely measured facts about who earns what, not as causal returns.

5.4 Regions

Table 5 and Figure 4 re-estimate the pooled equation within each macro-region. The gradient is steepest in the Nordeste, 10.2%, then the Sudeste, 9.9%, the Norte, 9.4%, the Centro-Oeste, 9.2%, and the Sul, 8.9%. The ranking follows the scarcity of educated workers across regions, and it matches what Duflo found across Indonesian districts, where returns were higher where initial schooling was lower (her Table 6).

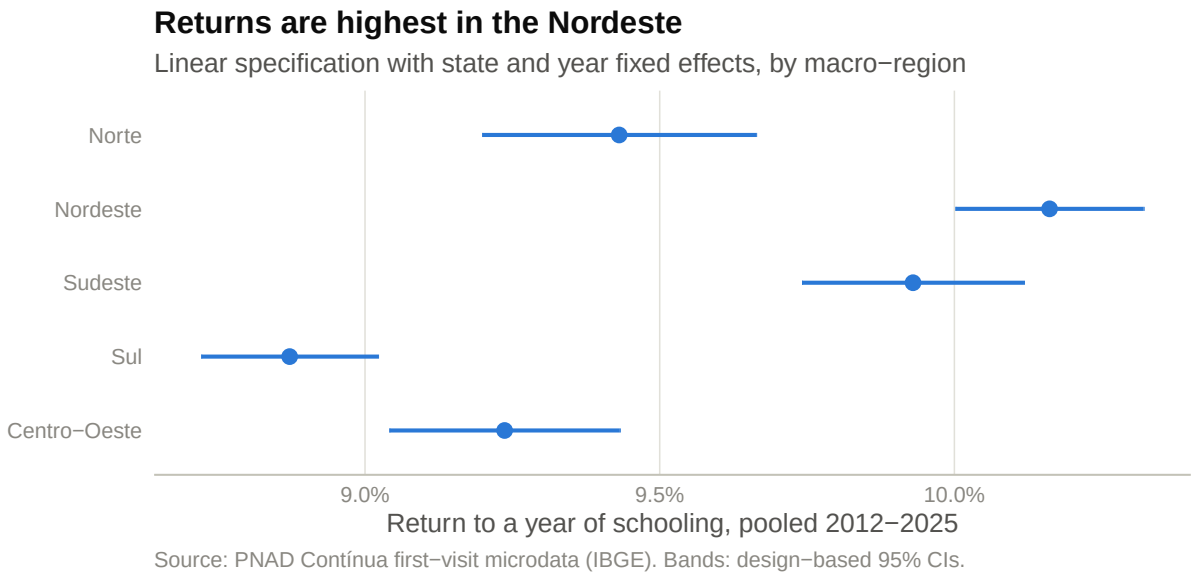


Figure 4: Return to a year of schooling by macro-region, pooled 2012–2025, linear specification with state and year fixed effects estimated separately by region, with design-based 95 percent confidence intervals.

Table 5: Return to a year of schooling by macro-region, pooled 2012-2025

Region	Estimate	SE	95% CI	Observations
Norte	0.0943	(0.0012)	[0.092, 0.097]	227,543
Nordeste	0.1016	(0.0008)	[0.100, 0.103]	505,528
Sudeste	0.0993	(0.0010)	[0.097, 0.101]	506,295
Sul	0.0887	(0.0008)	[0.087, 0.090]	336,062
Centro-Oeste	0.0924	(0.0010)	[0.090, 0.094]	198,139

Design-based standard errors in parentheses, computed from the PNAD Contínua strata, primary sampling units and first-visit weights (V1032). Dependent variable: log real hourly earnings in the main job (habitual monthly earnings at 2025 prices divided by 52/12 times weekly hours). Sample: occupied persons aged 25-64 with positive earnings and 1-120 weekly hours. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Linear specification with state and year fixed effects, estimated separately by region.

6 Robustness

Table 6 re-estimates the pooled coefficient under six sample and measurement variants and the two inference shortcuts. The point estimate is 9.76% at baseline; 10.02% with effective instead of habitual earnings; 10.19% for formal workers only; 9.89% for wage employees only, the sample closest to Duflo's; 9.54% when workers aged 18–24 are added; and 9.24% when the 0.5 percent tails of log earnings are dropped. The range, 9.2% to 10.2%, is narrow next to the differences across groups and levels, and each change goes the way one would expect: better-measured and formal earnings carry a slightly steeper gradient, and young workers, whose earnings are still

rising, a flatter one.

Table 6: Robustness of the pooled linear estimate

Variant	Years of schooling	SE	95% CI	Observations
(1) Baseline	0.0976	(0.0005)	[0.0967, 0.0986]	1,773,567
(2) Effective earnings	0.1002	(0.0005)	[0.0992, 0.1012]	1,728,784
(3) Formal only	0.1019	(0.0006)	[0.1007, 0.1030]	1,044,622
(4) Wage employees	0.0989	(0.0005)	[0.0979, 0.0999]	1,174,059
(5) Ages 18-64	0.0954	(0.0005)	[0.0945, 0.0963]	2,056,891
(6) Trimmed 0.5/99.5	0.0924	(0.0004)	[0.0916, 0.0933]	1,755,831
(7) Unweighted OLS, HC1	0.0936	(0.0001)	[0.0933, 0.0939]	1,773,567
(8) Weighted OLS, cluster UPA	0.0976	(0.0005)	[0.0966, 0.0987]	1,773,567

Coefficient on years of schooling in the linear specification with state and year fixed effects, pooled 2012-2025. Rows (1)-(6) use design-based inference (PNAD Contínua strata, PSUs and first-visit weights); (7) ignores weights and design and uses HC1 standard errors; (8) uses the weights with standard errors clustered by UPA but ignores strata. (2) uses effective instead of habitual earnings; (3) keeps signed-card or statutory employees and contributing self-employed; (4) keeps wage employees; (5) widens the age range; (6) drops the 0.5% tails of log earnings.

The inference comparison is the more instructive part of the table. Unweighted OLS with HC1 standard errors moves the point estimate to 9.36% and reports a standard error of 0.00015, 3.4 times smaller than the design-based 0.00049. The gap is the clustering of households within census tracts, which the robust standard error ignores. Weighted OLS with standard errors clustered by tract recovers the design-based point estimate exactly and a standard error of 0.00052, slightly above the design-based one because it forgoes the precision that stratification adds. For a coefficient this precisely estimated the practical consequence is small, but the same factor of three applies to every comparison across groups and years in the paper.

7 What the numbers cannot tell us

The coefficients in this paper measure how much more educated workers earn. They do not measure how much more a given person would earn after one more year of school, for the reasons set out by Griliches (1977) and surveyed by Card (1999): people who stay in school longer tend to have more ability and better-off families, both of which raise earnings on their own, and in Brazil Lam and Schoeni (1993) showed that family background accounts for a

sizeable share of the measured return. Earnings are also observed only for people who work, and who works is not random (Heckman 1979). Duflo's Indonesian estimates were designed to escape both problems, and even they turn out to be weakly identified under reanalysis. The design-based standard errors, however carefully computed, say nothing about these biases; they only make the descriptive estimate honest about its sampling precision.

Measurement matters for the magnitudes. Hourly earnings combine reported monthly earnings with reported weekly hours, often given by another household member, so errors in hours enter the dependent variable directly; trimming and the effective-earnings variant bound the influence of outliers, not of systematic misreporting. Years of schooling say nothing about school quality, which Behrman and Birdsall (1983) showed long ago matters as much as quantity in Brazil and which varies enormously across states and between public and private schools. The self-employed and employers are in the sample, and their earnings mix the return to labour with the return to capital; the wage-employee variant shows that this choice moves the pooled coefficient by little.

Two features of the data limit comparisons over time. Nothing can be said about 2020 and 2021. For 2022, the current microdata file gives mean labour income 3.3 percent above the values IBGE published for that year, while every other year matches within 0.01 percent; I use the microdata as released and record the discrepancy. All files are the 2025 revision of the series, re-weighted to post-Census population projections, and the published aggregates may be revised again.

Finally, the PNAD Contínua records where a worker lives, not where she was born or went to school. Interstate migration is selective, so the regional gradients describe where educated workers earn, not where they were educated.

8 Conclusion

Among occupied Brazilians aged 25–64, a year of schooling is associated with about 10% higher hourly earnings in every survey year between 2012 and 2025. The number is stable, but the relationship behind it is not. A completed degree is worth 1.45 log points over no schooling, the step from secondary to tertiary education carries an implied 20% a year, and every level premium fell over the period as the workforce moved up that convex profile. The gradient is steeper for women, flatter for Black and Brown workers and steepest in the Nordeste.

The paper also shows what careful descriptive work requires. Reproducing IBGE's published aggregates from the microdata caught the exact definitions of the official series and exposed a discrepancy in one year; stating the inference choices showed that the common shortcut of robust standard errors on unweighted data overstates precision by a factor of three. Those are the lessons of the reanalysis of Duflo's paper, applied before any causal claim is attempted.

The causal question remains open for Brazil. Answering it in Duflo's spirit requires a program with a documented allocation rule and data on where people were born, which points to the 1998 FUNDEF school-finance reform and the 2010 or 2022 Censuses rather than to the PNAD Contínua. The estimates here are the benchmark against which such a design should be judged.

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Appendix

Table 7: Sample construction: unweighted counts at each filter step

Year	All persons	Aged 25-64	Occupied	Positive earnings	Hours 1-120	Analysis sample
2012	446,445	224,410	153,037	146,662	146,662	146,662
2013	461,033	233,512	159,719	152,993	152,993	152,993
2014	465,038	239,141	163,589	156,830	156,830	156,830
2015	459,273	238,095	161,463	155,348	155,348	155,348
2016	459,718	239,934	159,229	154,616	154,616	154,607
2017	457,992	239,895	156,825	152,414	152,414	152,398
2018	452,654	238,772	155,788	151,588	151,588	151,572
2019	443,790	234,851	154,705	150,793	150,793	150,773
2022	376,821	201,191	133,465	130,693	130,693	130,659
2023	383,049	204,904	136,580	134,015	134,015	133,978
2024	381,815	205,223	139,788	137,298	137,298	137,271
2025	408,364	220,959	152,873	150,503	150,503	150,476

First-visit annual microdata, PNAD Contínua. Each column applies one more filter to the column to its left: aged 25-64; occupied in the reference week; positive habitual earnings in the main job; 1-120 weekly hours in the main job; schooling and race reported.

Table 8: Official aggregates reproduced from the survey design

Check	Estimate	Official	Rel. diff.	Status
2012 mean real labour income, 14+ occupied, habitual (R$ 2025)	3082.610	3083.000	0.013%	pass
2012 mean real labour income, 14+ occupied, effective (R$ 2025)	3114.480	3114.000	0.015%	pass

Check	Estimate	Official	Rel. diff.	Status
2013 mean real labour income, 14+ occupied, habitual (R$ 2025)	3184.870	3185.000	0.004%	pass
2013 mean real labour income, 14+ occupied, effective (R$ 2025)	3210.320	3210.000	0.010%	pass
2014 mean real labour income, 14+ occupied, habitual (R$ 2025)	3279.930	3280.000	0.002%	pass
2014 mean real labour income, 14+ occupied, effective (R$ 2025)	3303.460	3303.000	0.014%	pass
2015 mean real labour income, 14+ occupied, habitual (R$ 2025)	3151.410	3151.000	0.013%	pass
2015 mean real labour income, 14+ occupied, effective (R$ 2025)	3194.440	3194.000	0.014%	pass
2016 mean real labour income, 14+ occupied, habitual (R$ 2025)	3163.870	3164.000	0.004%	pass
2016 mean real labour income, 14+ occupied, effective (R$ 2025)	3295.310	3295.000	0.010%	pass
2016 mean years of schooling, 25+, Total	9.120	9.100	0.210%	pass
2016 mean years of schooling, 25+, Homens	8.930	8.900	0.327%	pass
2016 mean years of schooling, 25+, Mulheres	9.290	9.200	1.002%	pass
2017 mean real labour income, 14+ occupied, habitual (R$ 2025)	3147.430	3147.000	0.014%	pass
2017 mean real labour income, 14+ occupied, effective (R$ 2025)	3254.120	3254.000	0.004%	pass
2017 mean years of schooling, 25+, Total	9.240	9.200	0.466%	pass
2017 mean years of schooling, 25+, Homens	9.050	9.000	0.575%	pass
2017 mean years of schooling, 25+, Mulheres	9.420	9.400	0.179%	pass
2018 mean real labour income, 14+ occupied, habitual (R$ 2025)	3216.720	3217.000	0.009%	pass
2018 mean real labour income, 14+ occupied, effective (R$ 2025)	3343.950	3344.000	0.002%	pass
2018 mean years of schooling, 25+, Total	9.430	9.400	0.287%	pass
2018 mean years of schooling, 25+, Homens	9.220	9.200	0.211%	pass
2018 mean years of schooling, 25+, Mulheres	9.620	9.600	0.166%	pass
2019 mean real labour income, 14+ occupied, habitual (R$ 2025)	3204.130	3204.000	0.004%	pass
2019 mean real labour income, 14+ occupied, effective (R$ 2025)	3325.570	3326.000	0.013%	pass

Check	Estimate	Official	Rel. diff.	Status
2019 mean years of schooling, 25+, Total	9.530	9.500	0.298%	pass
2019 mean years of schooling, 25+, Homens	9.310	9.300	0.054%	pass
2019 mean years of schooling, 25+, Mulheres	9.730	9.700	0.324%	pass
2022 mean real labour income, 14+ occupied, habitual (R$ 2025)	3134.070	3032.000	3.366%	documented exception
2022 mean real labour income, 14+ occupied, effective (R$ 2025)	3254.660	3150.000	3.323%	documented exception
2022 mean years of schooling, 25+, Total	9.830	9.800	0.263%	pass
2022 mean years of schooling, 25+, Homens	9.610	9.600	0.088%	pass
2022 mean years of schooling, 25+, Mulheres	10.020	10.000	0.229%	pass
2023 mean real labour income, 14+ occupied, habitual (R$ 2025)	3249.040	3249.000	0.001%	pass
2023 mean real labour income, 14+ occupied, effective (R$ 2025)	3371.440	3371.000	0.013%	pass
2023 mean years of schooling, 25+, Total	9.930	9.900	0.326%	pass
2023 mean years of schooling, 25+, Homens	9.730	9.700	0.322%	pass
2023 mean years of schooling, 25+, Mulheres	10.110	10.100	0.145%	pass
2024 mean real labour income, 14+ occupied, habitual (R$ 2025)	3368.090	3368.000	0.003%	pass
2024 mean real labour income, 14+ occupied, effective (R$ 2025)	3492.080	3492.000	0.002%	pass
2024 mean years of schooling, 25+, Total	10.050	10.100	0.482%	pass
2024 mean years of schooling, 25+, Homens	9.810	9.900	0.878%	pass
2024 mean years of schooling, 25+, Mulheres	10.270	10.300	0.316%	pass
2025 mean real labour income, 14+ occupied, habitual (R$ 2025)	3559.530	3560.000	0.013%	pass
2025 mean real labour income, 14+ occupied, effective (R$ 2025)	3694.130	3694.000	0.003%	pass
2025 mean years of schooling, 25+, Total	10.240	10.200	0.386%	pass
2025 mean years of schooling, 25+, Homens	10.010	10.000	0.139%	pass
2025 mean years of schooling, 25+, Mulheres	10.440	10.400	0.422%	pass

Estimate: design-based mean from the first-visit microdata (weights V1032, strata, PSUs). Official: IBGE/SIDRA tables 7445 (labour income, same microdata and deflator) and 7126 (years of schooling, second-quarter supplement, one decimal).